AI Usage and Teamwork Report

Team EV · CptS 484 Project THEIA · Phase I · October 18, 2026

Owner: Zichuo Wang · Co-author: Young Kim

1. Summary

Team EV used generative AI on every Phase I work product: to generate candidate scenarios, draft document templates, critique drafts, draw mock-up screens, and help assemble and cross-check the final submission. The team's rule (success criterion SC-5 in the Project Plan) was that no AI output enters a deliverable until its owner has checked it against the course documents and a second member has reviewed it.

This report lists the tools used (§2), how each work product used them (§3), the review process (§4), and concrete cases where AI output was rejected or corrected (§5). §6 covers how the team worked together. The evidence (screenshots and logs) is in Appendix A.

The record is uneven. James Zuniga kept a detailed log; Zichuo Wang, Young Kim, and Austin Aragon posted shorter notes in Discord; Justus Jones added a log of his Claude Code sessions on Oct 18, after this report was first assembled.

2. Tools Used

Member	Tool(s)	Used for	Work product
James Zuniga	Gemini, NotebookLM	Turning the team's issues list into AS-IS/TO-BE scenario candidates; comparing scenario strengths to pick the Top Priority; applying review feedback to the slides	D-1 slides
Austin Aragon	Gemini	Reviewing and critiquing the WRS §2 draft	D-2 WRS §2
Young Kim	ChatGPT; Claude	Checking which requirements could be shown in a mock-up; drawing the 10 mock-up screen images	D-4 prototype
Zichuo Wang	Claude Code	Weekly check-in messages; review feedback on all work products; WRS templates (front matter, §2, §3); final assembly, consistency check, plan and report drafting; comparing Justus Jones's §3 draft with the assembled WRS and merging it (Oct 18)	D-2, D-3, D-5, D-6, assembly
Justus Jones	Claude Code (Claude Opus 5.5)	Orienting in the plan and repo; drafting an independent WRS §3 from WRS §2; checking that draft against the course documents	D-3 WRS §3 (draft, Oct 18)

3. How the Team Used AI, by Work Product

D-1 AS-IS/TO-BE slides (James Zuniga). On Oct 4 James gave Gemini/NotebookLM the team's issues list from Discord and asked for help building scenario pairs. The first exchange produced 11 candidates, all rejected (they restated his request instead of using the issues). With the issues list pasted in, it produced 4 scenario candidates; he kept 2 and rejected 2, then chose the three pairs that best covered the issues.

He then asked it to list the strengths of each pair and used that list to choose the Top Priority scenario (off-route recovery and emergency escalation). On Oct 15 he used Gemini to turn Zichuo's review feedback into a list of slide changes, and then rewrote the wording in his own words, rebuilt the layout, and added animations himself.

D-2 WRS §2 Issues (Austin Aragon, James Zuniga). The issues came from the members' own elicitation (`Issues.md`). Zichuo turned the consolidated list into an AI-assisted §2 template on Oct 14, with options, decisions, and rationales marked as drafts. Austin used Gemini to critique the draft on Oct 16 and rewrote the rationale for five issues (DI1, DI4, FRI2, FRI4, NFRI4) in his own terms.

D-3 WRS §3 Improved Understanding (Justus Jones, Zichuo Wang). Zichuo drafted §3 (Problems, Goals, objectives, FR/NFR with specifications, traceability) with AI help on Oct 14 as a template for the §3 owner, labeled as AI-generated. Justus Jones, the owner, did not reply during the final week, so Zichuo completed §3 at assembly, checking every requirement against §2, the recorded slides, and the prototype. On Oct 18 Justus used Claude Code to write an independent §3 from Austin's §2 (WRS-Section2-Updated.md), deliberately without the assembled version, and then had it check the draft against the World Assumptions Masterlist, the Requirements Review Checklist, the Preliminary Definition, the project specification, and the sample WRS. Zichuo compared his draft with the assembled §3 using Claude Code, and the team merged its additions into WRS v1.1 (§5, items 6 and 7).

D-4 Prototype and User Manual (Young Kim, Zichuo Wang). On Oct 1 Young asked ChatGPT to look over the requirements and flag anything that could not be mocked up; it also suggested ideas for customizing settings. On Oct 15 Young used Claude to draw 10 screen images (phone screen, speech bubbles, and a "Feel / Do" box per screen), which became the slides of the mock-up. The manual was written by Young and revised to v1.1 at assembly.

D-5 Project Plan (Zichuo Wang, Young Kim). AI helped draft the weekly check-ins (e.g., the Oct 1 Week 3 check-in) and the final plan update. The schedule, roles, and reassignment decisions were made by the team liaison.

4. Review and Evaluation Process

The rule (SC-5). Every AI-assisted passage is checked by the owner of the work product against the source documents, and by one other member, before it ships. "Checked" means: every claim traced to the Preliminary Definition, the World Assumptions Masterlist, or a team decision; nothing invented (numbers, features, stakeholders); and the wording rewritten so it reads as the team's own.

Source documents used for checking: THEIA Preliminary Definition v1.0, World Assumptions Masterlist v1.0, Project Specification Phase I v2.0, the WRS template and sample, and the Requirements Review Checklist.

Work product	Owner check	Second review
D-1 slides	James Zuniga: kept 2 of 4 scenario candidates; picked the Top Priority himself; rewrote wording	Zichuo Wang's Oct 14 feedback (checked against the spec, the WRS template, and the Masterlist); James applied it Oct 15
D-2 WRS §2	Austin Aragon: critique with Gemini, five rationales rewritten	Zichuo Wang at assembly: plain-language edits, typo fixes

Work product	Owner check	Second review
D-3 WRS §3	Zichuo Wang (co-author): assembled §3 labeled AI-generated, every item traced to a §2 issue. Justus Jones (owner): his Oct 18 draft checked against the Masterlist, Review Checklist, Preliminary Definition, spec, and sample WRS	Young Kim (review); cross-check against slides and prototype at assembly; Justus Jones's draft checked against the same before merging
D-4 prototype, manual	Young Kim: screens checked against the TO-BE scenarios	Zichuo Wang at assembly: screens and manual checked against WRS §3
D-5 plan, D-6 report	Zichuo Wang	Team review before submission

Decisions people made. AI produced candidates and checks. The team made the decisions, including:

- the scope calls (indoor connectors only; static obstacles only; public pass-through rooms; the sensor list);
- which issues to keep, and the decision and rationale for each;
- the three-plus-one scenario pairs and the Top Priority choice (James Zuniga);
- at assembly, how to resolve conflicts between work products (Zichuo Wang): the recorded slides are fixed, the WRS follows the slides, and the prototype and manual follow the WRS; one warning distance (5 steps) everywhere; the role changes; and which parts of Justus Jones's late §3 draft to merge.

5. What the Team Caught and Corrected

1. Scenario candidates that ignored the issues (D-1, Oct 4). The first 11 candidates from Gemini/NotebookLM were all rejected because they restated the request rather than working from the issues list. Of the next 4 candidates, 2 were rejected. (Appendix A, Figures A6–A7.)

2. Out-of-scope features in the slides (D-1, Oct 14–15). The AI-drafted scenarios included features the Masterlist rules out. Review feedback caught them and James changed the slides: "dynamic crowd" detection in Scenario 4 was replaced by "avoids areas the map marks as busy" (moving obstacles are out of scope, WA5), and "3D spatial audio" on the comparison slide was replaced by "hands-free voice + vibration guidance". Name errors ("Stevie calls caretaker Bella") were also fixed. (Figures A17–A18.)

3. A habit-learning assumption (D-1 / D-2). The AI's Elena scenario had THEIA "check Elena's calendar and path history". The team's decision in FRI1 is to use an entered schedule only, without learning from location history. The final slide does not use path history. (Figure A11.)

4. Generic rationale in §2 (D-2, Oct 16). Austin replaced five AI-drafted rationales with his own. At assembly, the rewritten rationales were edited into plain language and a typo ("usering") was fixed. The decisions themselves did not change.

5. Conflicts between work products (assembly, Oct 17). The cross-check found, among others:

- the user manual's gesture and vibration tables were copies of its voice-command table;
- the manual said Comfortable mode "ignores congestion", while the WRS and Scenario 4 say it avoids busy areas;
- the WRS gave three different turn-warning distances, and disagreed with itself on who sets the default mode;

- the prototype listed “two-finger tap” for “Where am I?” while its own screens showed a two-finger double-tap;
- the manual described a caretaker-reply feature with no requirement behind it.

Each was fixed in the direction the team set. The AI found the conflicts; the resolution rules were the liaison’s.

6. Justus Jones’s own checks (D-3, Oct 18). Justus’s log records that the AI’s first role list came from the repository README rather than the plan, so he corrected it; that the Masterlist check found a world assumption contradicting it (users “use” a cane or dog, where the Masterlist says they “might”) and four missing assumptions, which he fixed; and that he rejected a suggested diagram and reverted README edits he had not asked for. (Appendix A.5.)

7. A late §3 draft that conflicted with the recorded scenarios (D-3, Oct 18). Justus’s §3 was built from §2 alone, before the slides, prototype, and manual were reconciled. Checked against them, it conflicted in 11 places, among others: the command “Blocked” instead of “This way is blocked”; an off-route threshold of 15 s or 5 m (room-level beacons cannot measure 5 m) instead of 30 s; “Help” texting at once instead of the four-option prompt on prototype Screen 4; Comfortable mode allowing 50% extra time and Safe mode avoiding crowds, against Scenario 4; no arrival requirement, although the prototype and manual use one; and a prototype mapping based on the mock-up before its fixes (“two-finger tap”). These parts were not adopted. His priorities, NFR acceptance criteria, error handling (no route, wrong floor, no cellular signal), and Preliminary Definition coverage table were merged, rewritten to the WRS’s IDs and numbers.

6. Teamwork

Organization. Five members, one owner and one co-author per work product, everyone reviewing. Discord was the main channel (one channel per work product), with weekly written checkpoints on Sundays posted by the liaison. Drafts lived in Google Drive; submitted versions are in the GitHub repository.

Who did what.

Member	Contribution to Phase I
Zichuo Wang	Liaison; Project Plan (preliminary and final); weekly check-ins and review feedback; WRS templates; WRS §3 co-author (completed §3 at assembly); prototype co-author; this report; final assembly and consistency check
James Zuniga	AS-IS/TO-BE slides and the recorded presentation; Top Priority analysis; WRS §2 co-author
Austin Aragon	WRS §2 Issues: consolidated issues list, options, decisions, rationale; slides co-author
Young Kim	Prototype mock-up (5 screens, 10 images) and preliminary user manual; Project Plan co-author; this report’s co-author
Justus Jones	WRS §3 owner. Did not reply to the team during the final week; on Oct 18 submitted an independent §3 draft, whose priorities, acceptance criteria, error handling, and Preliminary Definition coverage were merged into WRS v1.1, and his AI usage log

What worked. The co-author model limited the damage when an owner went silent: §3 already had a co-author and a draft, so it was completed in the last week without losing a work product, and the owner’s late draft could still be reviewed and merged. The Oct 14 feedback round, with fixed deadlines (Thu Oct 15 fixes, Fri Oct 16 video, Sat Oct 17 assembly, Sun Oct 18 submission), held.

What we would change. The WRS §2 internal deadline (Oct 4) slipped to Oct 16, and AI evidence was collected at the end instead of continuously, which is why parts of this record are thin. The D-3 owner's draft arrived after assembly, so it was merged on the deadline day instead of being reviewed at a weekly checkpoint.

Appendix A. Evidence

A.1 Zichuo Wang

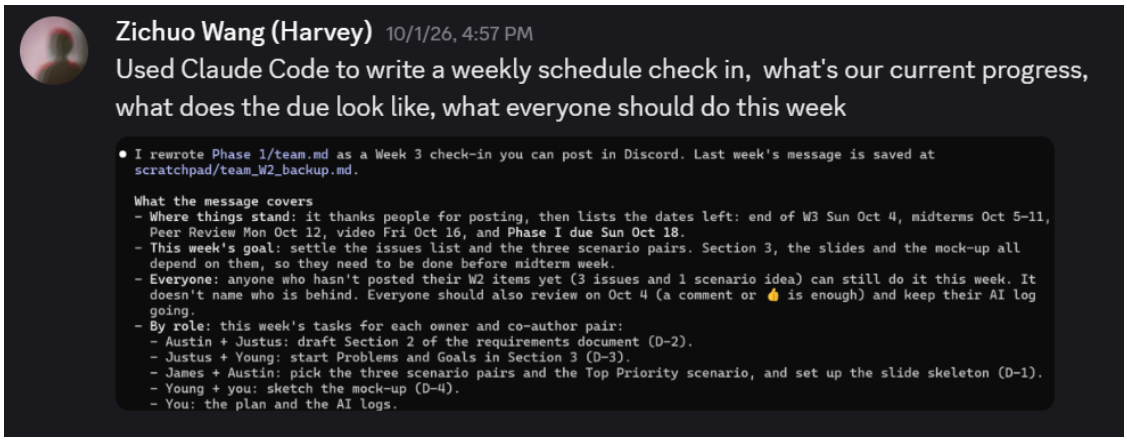


Figure A1. Oct 1: Week 3 check-in drafted with Claude Code.

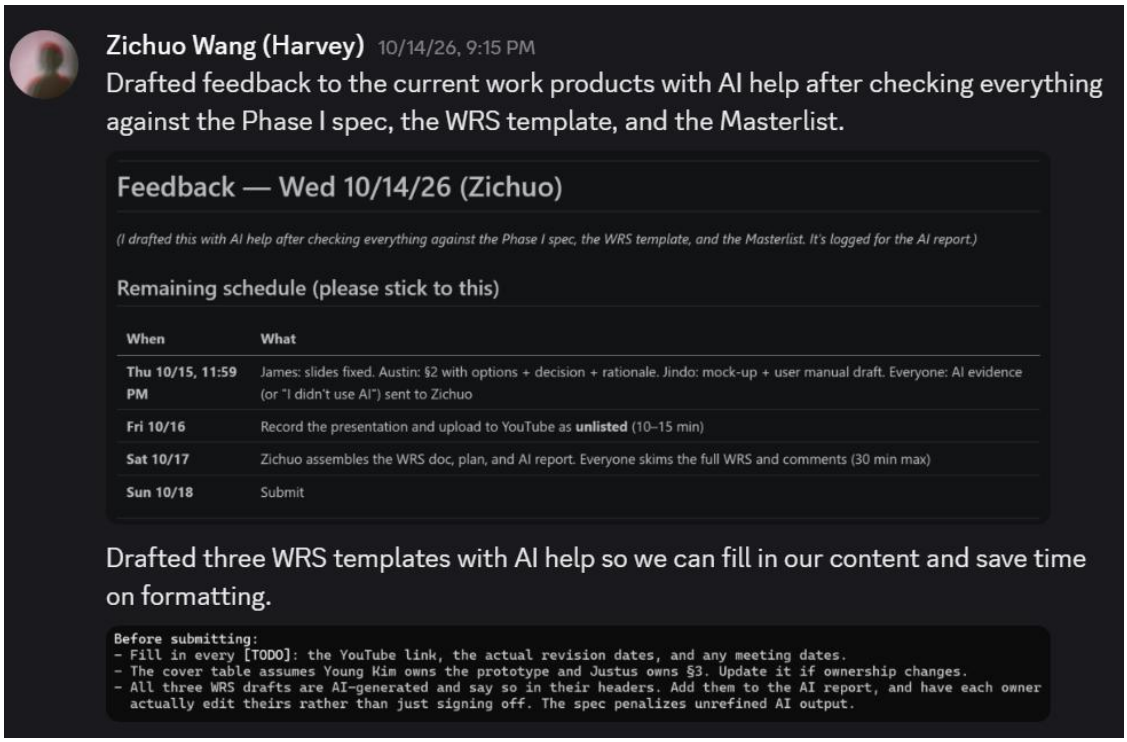


Figure A2. Oct 14: review feedback drafted with AI after checking every work product against the Phase I spec, WRS template, and Masterlist; WRS templates labeled as AI-generated with [TODO]s.

A.2 Young Kim

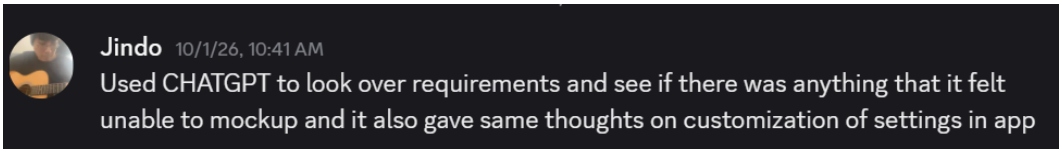


Figure A3. Oct 1: ChatGPT used to check which requirements could be mocked up.

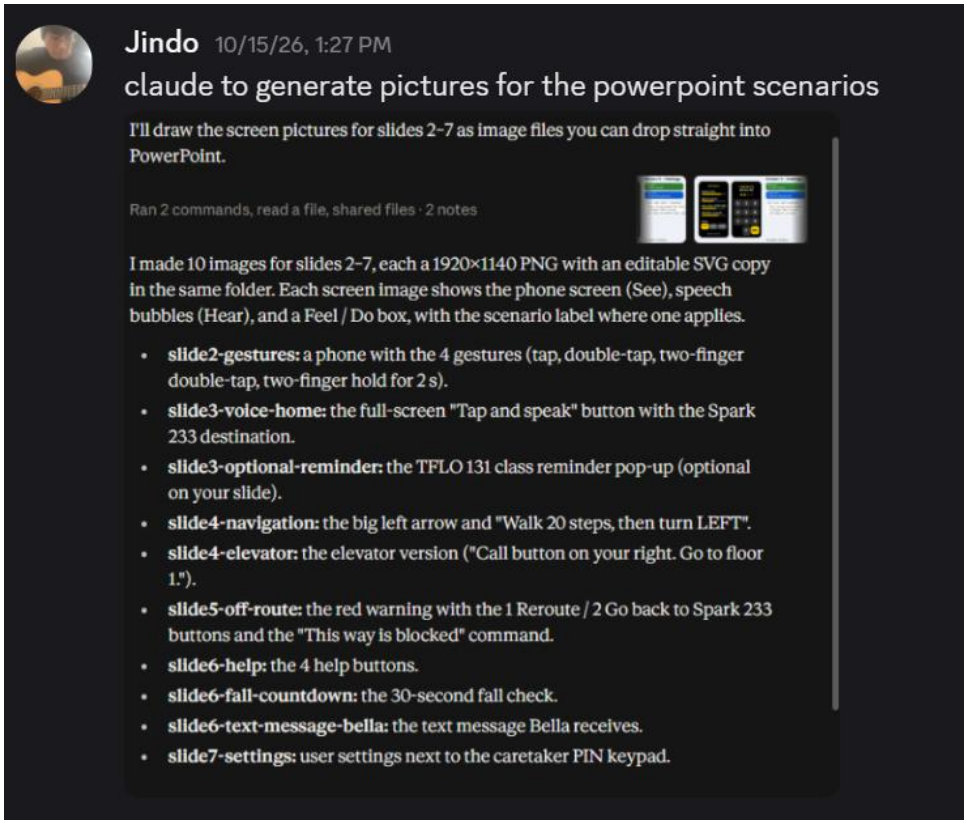


Figure A4. Oct 15: Claude used to draw the mock-up screen images.

A.3 Austin Aragon

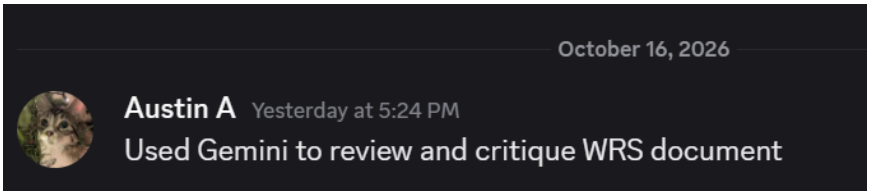
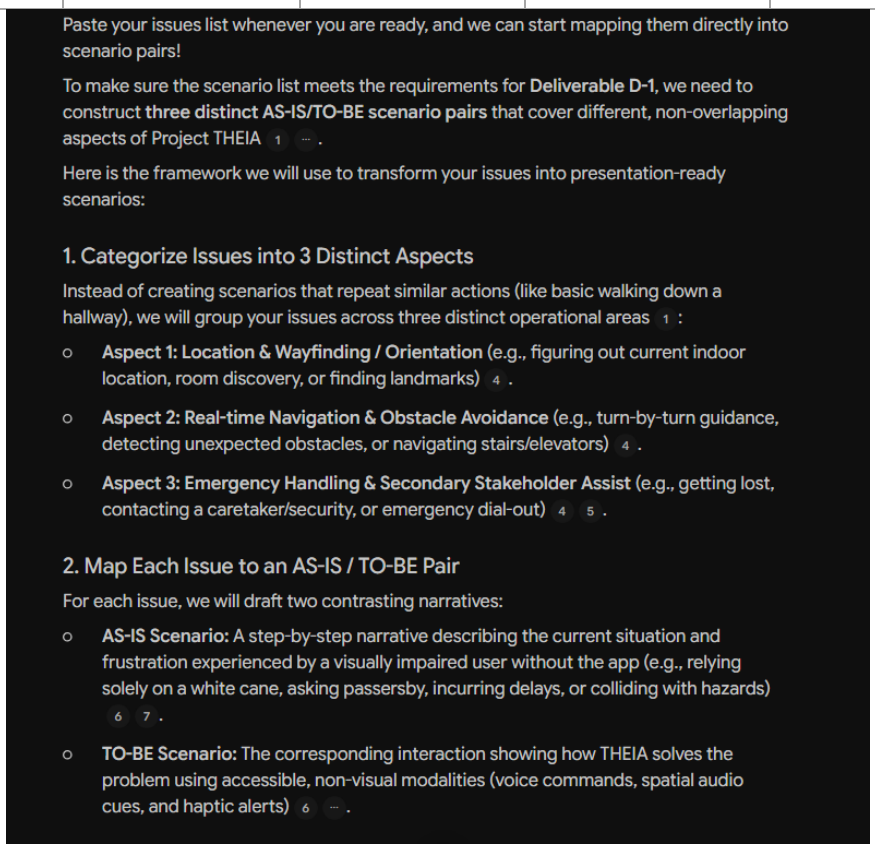


Figure A5. Oct 16: Gemini used to review and critique the WRS §2 draft.

A.4 James Zuniga

From *AI Usage Report V1.2* (James Zuniga), which also contains the full log text.

Date	Time	Asked for	Came back	Changed by James	Verified against
Oct 4	~40 min	Help building AS-IS/TO-BE scenarios	11 candidates, 11 rejected (confirmed the request, waited for input)	Supplied the issues list from Discord	Issues list (Discord)
Oct 4	—	Scenarios from the pasted issues list	4 candidates, 2 accepted, 2 rejected	Chose the pairs that best fit the issues	Issues list (Discord)
Oct 4	~3 min	Strengths of each scenario pair for the Top Priority choice	3 candidates, 1 accepted, 2 rejected	Decided the Top Priority scenario himself	Issues list (Discord)
Oct 15	~20 min	Changes needed to apply the review feedback	9 changes, all accepted	Reworded in his own words, rebuilt layout, added animations	Feedback image from Zichuo Wang (#as-is-to-be)



Paste your issues list whenever you are ready, and we can start mapping them directly into scenario pairs!

To make sure the scenario list meets the requirements for **Deliverable D-1**, we need to construct **three distinct AS-IS/TO-BE scenario pairs** that cover different, non-overlapping aspects of Project THEIA [1] [...].

Here is the framework we will use to transform your issues into presentation-ready scenarios:

1. Categorize Issues into 3 Distinct Aspects

Instead of creating scenarios that repeat similar actions (like basic walking down a hallway), we will group your issues across three distinct operational areas [1]:

- **Aspect 1: Location & Wayfinding / Orientation** (e.g., figuring out current indoor location, room discovery, or finding landmarks) [4].
- **Aspect 2: Real-time Navigation & Obstacle Avoidance** (e.g., turn-by-turn guidance, detecting unexpected obstacles, or navigating stairs/elevators) [4].
- **Aspect 3: Emergency Handling & Secondary Stakeholder Assist** (e.g., getting lost, contacting a caretaker/security, or emergency dial-out) [4] [5].

2. Map Each Issue to an AS-IS / TO-BE Pair

For each issue, we will draft two contrasting narratives:

- **AS-IS Scenario:** A step-by-step narrative describing the current situation and frustration experienced by a visually impaired user without the app (e.g., relying solely on a white cane, asking passersby, incurring delays, or colliding with hazards) [6] [7].
- **TO-BE Scenario:** The corresponding interaction showing how THEIA solves the problem using accessible, non-visual modalities (voice commands, spatial audio cues, and haptic alerts) [6] [...].

Figure A6. Oct 4: scenario framework proposed by the AI.

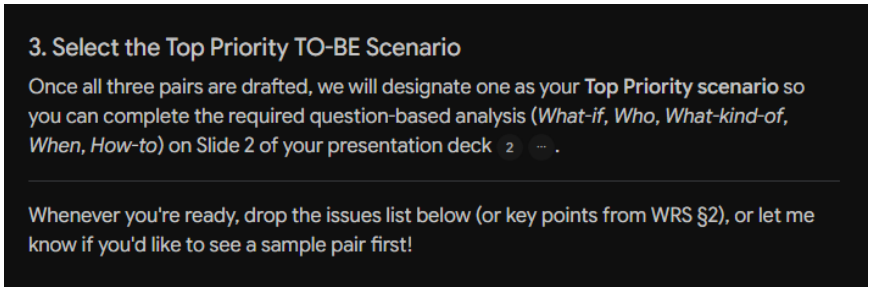


Figure A7. Oct 4: Top Priority selection step.

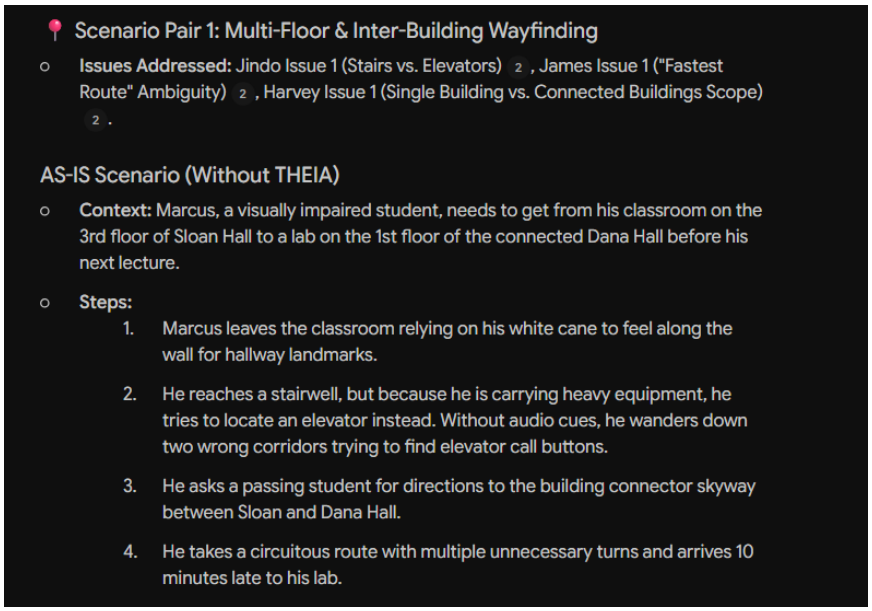


Figure A8. AI draft, Scenario Pair 1 AS-IS (Marcus).

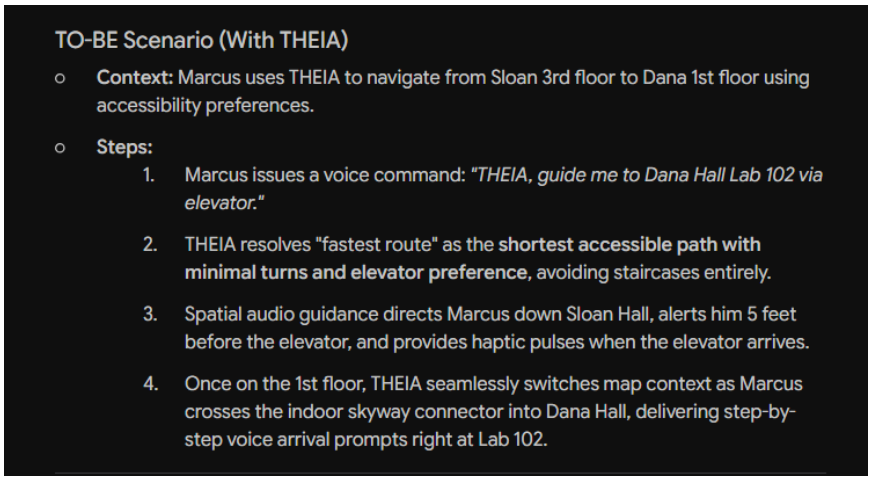


Figure A9. AI draft, Scenario Pair 1 TO-BE.

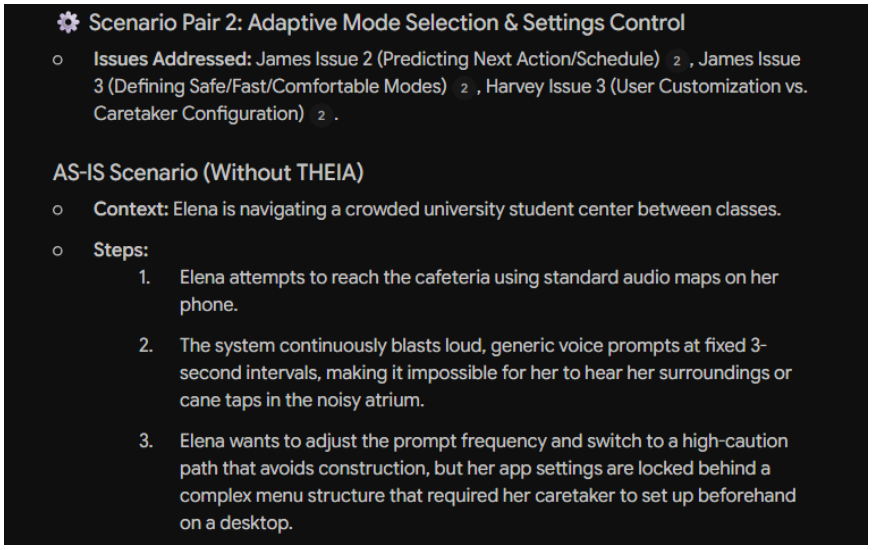


Figure A10. AI draft, Scenario Pair 2 AS-IS (Elena).

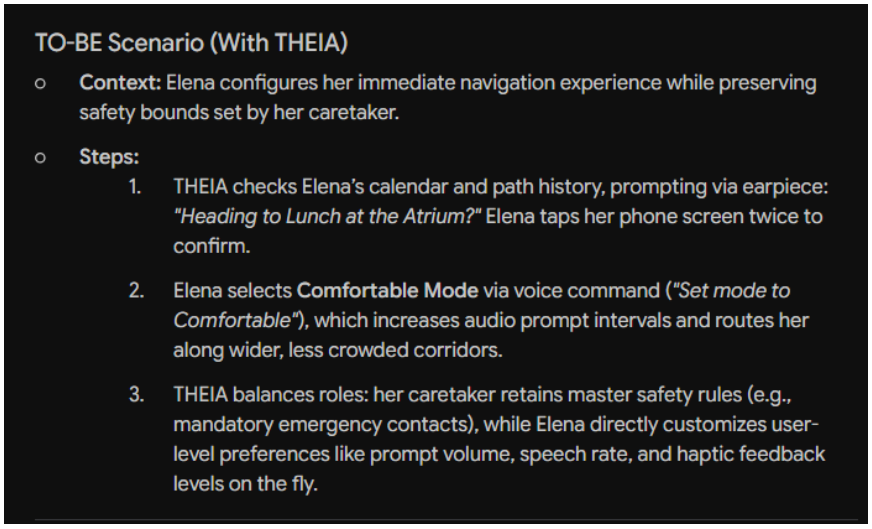


Figure A11. AI draft, Scenario Pair 2 TO-BE: "checks Elena's calendar and path history" (not used).

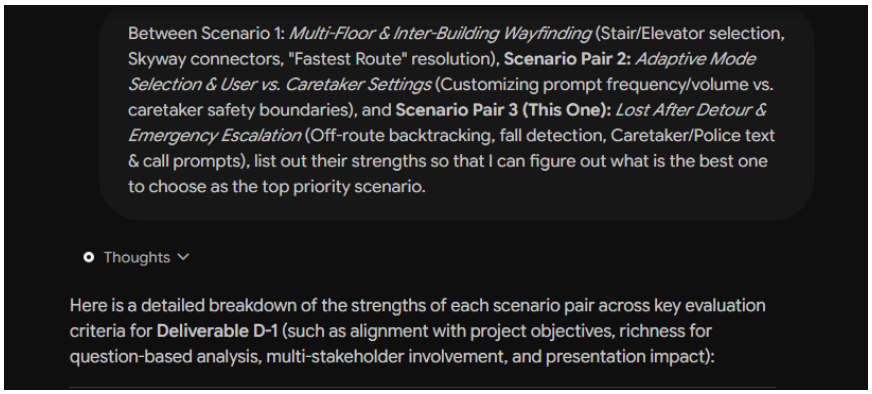


Figure A12. Oct 4: request to compare the scenario pairs.

1. Scenario Pair 1: Multi-Floor & Inter-Building Wayfinding

(Stair/Elevator selection, Skyway connectors, "Fastest Route" resolution)

- **Core Vision Alignment:** Represents the **primary functional objective** of Project THEIA—navigating an indoor domain across multiple floors and connected buildings [1].
- **Strengths:**
  - **Directly Solves Core Functional Ambiguities:** Resolves major issues in WRS §2 regarding vertical transit (stairs vs. elevators) and defining what "fastest route" means (e.g., shortest accessible path vs. least turns) [2].
  - **Clear Sensor & Technical Focus:** Highlights physical infrastructure interaction—Bluetooth beacon handoffs, elevator arrival detection, and indoor map context switching.
  - **High Prototyping Feasibility:** Highly straightforward to demonstrate in non-visual UI mock-ups (e.g., voice prompts for "Fastest via Elevator").
- **Best Used For:** Proving that the system handles everyday, routine navigation tasks seamlessly.

Figure A13. Strengths, Scenario Pair 1.

2. Scenario Pair 2: Adaptive Mode Selection & Settings Control

(Customizing prompt frequency/volume vs. caretaker safety boundaries)

- **Core Vision Alignment:** Targets **Non-Functional Requirements (NFRs)**—usability, user-friendliness, and audio/haptic prompt pacing in loud environments [1].
- **Strengths:**
  - **Solves Operational & Customization Conflicts:** Addresses the tension between user autonomy (customizing speech rate and prompt intervals) and caretaker control (setting master emergency contacts and safety bounds).
  - **Strong Accessibility Focus:** Directly addresses the non-visual UI experience by showing how audio cues adjust dynamically to crowded, noisy environments rather than blasting fixed-interval voice commands.
- **Best Used For:** Demonstrating how THEIA respects user accessibility preferences while maintaining safety constraints.

Figure A14. Strengths, Scenario Pair 2.

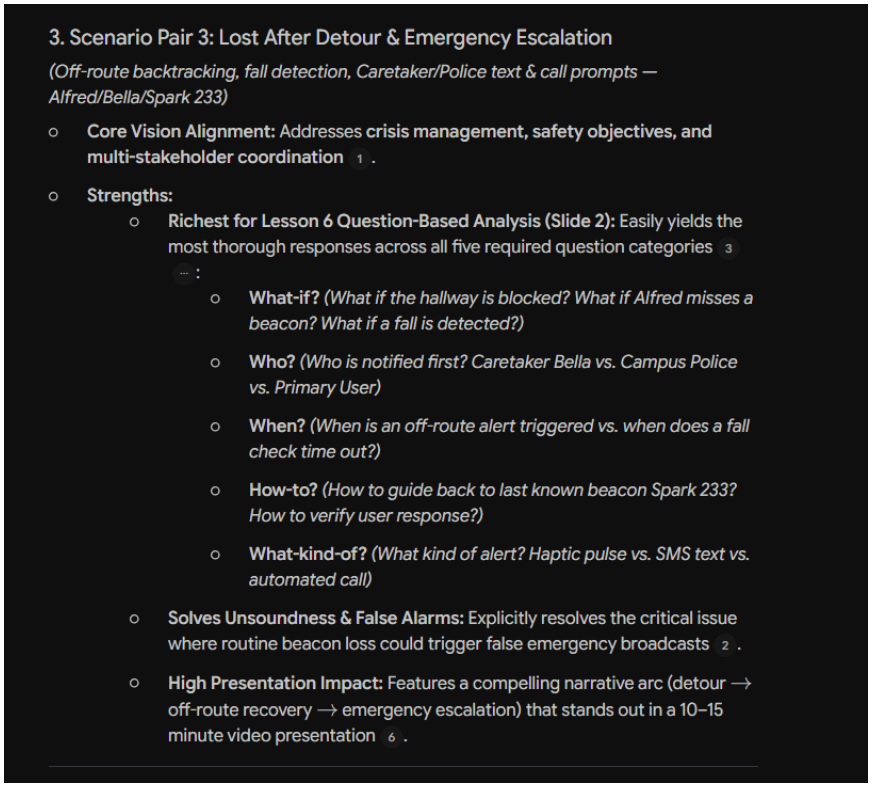


Figure A15. Strengths, Scenario Pair 3 (chosen as Top Priority).

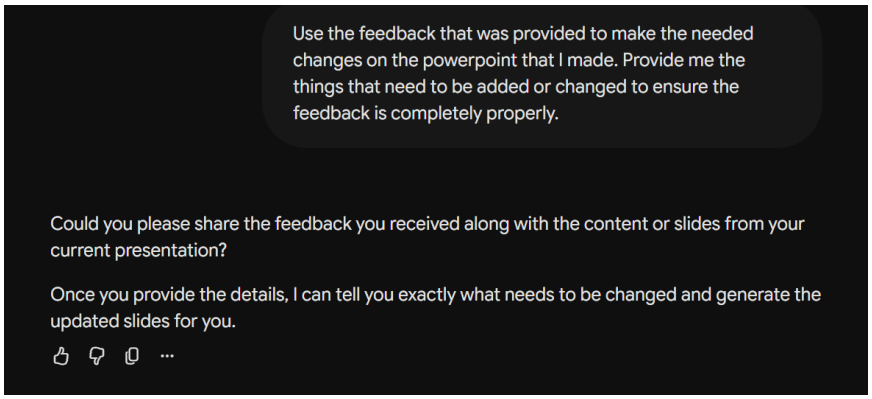


Figure A16. Oct 15: request to apply the review feedback.

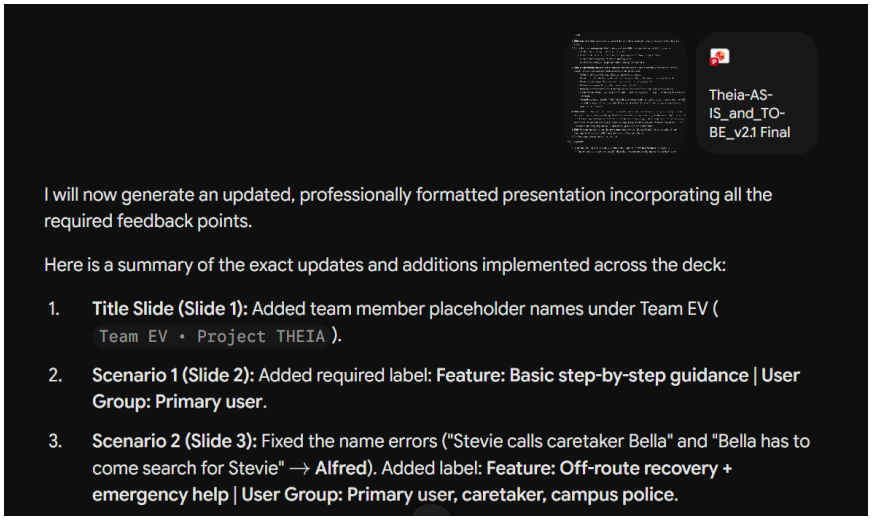


Figure A17. Oct 15: changes from the feedback (slides 1–3).

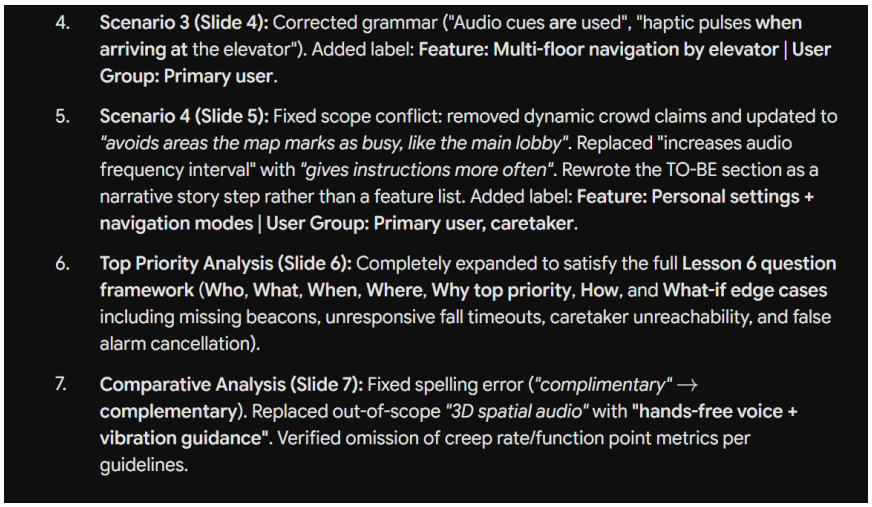


Figure A18. Oct 15: changes from the feedback (slides 4–7), including the removed out-of-scope claims.

A.5 Justus Jones

From Justus Jones's AI_Usage_Log.md (Jones_Justus_AI_Usage_Report folder), added Oct 18. Tool in every session: Claude Code (Claude Opus 5.5). Quoted as written, except that two fields of the third session, which the log leaves unfilled ("What I changed" and the check of the FR9, FR11, and FR12 rows against the prototype), are left out.

Oct 18 · Repo orientation and D-3 scope.

What I asked for: a read-through of the Preliminary Project Plan and the repo, then which work products were mine (D-3 owner, D-5 secondary per plan §2.3) and how D-3 depends on the others.

What came back: a summary of the project, the WRS (§2 Issues = D-2, §3 Improved Understanding = D-3), and the dependencies D-2 → D-3 → D-4/D-1/D-5/D-6. Useful: the summary and dependencies. Rejected: the first role list it gave, which came from the repo README rather than plan §2.3. I supplied the §2.3 table to correct it.

What I changed: confirmed roles against plan §2.3 myself.

Verified against: Team_EV_Preliminary_Project_Plan.docx §2.3; 03-wrs/README.md.

Oct 18 · WRS §3 draft (D-3).

What I asked for: a fresh WRS §3 for D-3, built only from Austin's §2 (WRS-Section2-Updated.md), without using the D-3 version Zichuo submitted on my behalf. I checked out the commit before it (fc0a30e) so that version wasn't in the working tree.

What came back: one draft with Problems P1–P8, Goals G1–G8, domain (ID1–ID3), stakeholders, world assumptions, objectives FO1–FO10/NFO1–NFO7, FR1–FR13, NFR1–NFR7, and three traceability tables. It kept the IDs §2 already cites in its "Satisfied by" rows. Useful: the structure and traceability. Problems found later (see next entry): one world assumption contradicted the Masterlist, four Masterlist assumptions were missing, and there were no priorities or error behavior.

What I changed: renamed it to WRS-Section3-Updated.md and set Austin Aragon as co-author.

Verified against: WRS §2 "Satisfied by" rows; plan §2.3.

Oct 18 · Checking §3 against course documents.

What I asked for: check the draft against the course documents from Canvas, one at a time, and fix what didn't match.

What came back:

- World Assumptions Masterlist v1.0: the draft said users use a cane or dog; the Masterlist says they might. Fixed. Added the four missing assumptions (English, smartphone accessibility experience, emergency dial-out, room-to-room only) and a Source column marking each assumption as Masterlist or Added.
- Requirements Review Checklist: added a priority for every requirement, error behavior for FR1/FR2/FR3/FR10, and removed a design detail (a specific gesture) from FR1.
- Preliminary Definition v1.0: mapped every statement to §3 (new table §3.3.4). FR10 only texted the caretaker though the Definition says "calls and messages"; fixed.
- Project Specification I v2.0 and the sample WRS: §3 covers everything §III.2.2 asks for. Added §3.3.5 mapping each FR to the prototype, as the sample does, and the facing direction to FR1 to match the prototype's "Where am I?".

Rejected: a suggestion to add a diagram; neither the spec nor our plan requires one.

Verified against: Preliminary Definition v1.0; World Assumptions Masterlist v1.0; Requirements Review Checklist; Project Specification I v2.0 §III.2.2; 484 - WRS.pdf (sample); Theia Prototype Mock-Up.

Oct 18 · Repo housekeeping and recovery.

What I asked for: role fixes in the 03-wrs files to match plan §2.3, this log, git guidance, and later a rebuild of my files after I accidentally deleted my local clone.

What came back: edits to 03-wrs/README.md and the §3 header. It also edited the other folders' READMEs, which I hadn't asked for. Rejected: those extra edits, which I reverted. After the deletion it

rebuilt §3 and this log from our conversation in my fork; I re-checked IDs and traceability in the rebuilt file.

What I changed: kept only the 03-wrs changes. I did all git operations myself; the AI made no commits.

Verified against: plan §2.3.